Sensor-Based IoT Simulation Using Tinkercad

Aim: Simulate a temperature sensor controlling an LED using Arduino.

Components: Arduino Uno, TMP36 sensor, LED, 220Ω resistor, breadboard, jumper wires.

Working: Arduino reads the TMP36 sensor. If temperature exceeds 30°C, LED turns ON; otherwise it remains OFF.

Arduino Code (summary):
const int sensorPin=A0; const int ledPin=13; Read analog value, convert to voltage and temperature, compare with 30°C, control LED.

Code Explanation: analogRead() reads the sensor, the reading is converted to voltage and temperature, digitalWrite() turns the LED ON/OFF based on the threshold.

Screenshots: Insert your Tinkercad circuit screenshot, LED OFF, LED ON, and Serial Monitor screenshots here before submission.

Applications: Smart homes, agriculture, healthcare monitoring, industrial automation, fire detection.

Conclusion: The simulation demonstrates a basic IoT monitoring system using a temperature sensor and Arduino.

References:
1. Arduino Documentation: https://docs.arduino.cc/
2. Tinkercad Circuits: https://www.tinkercad.com/circuits
3. Analog Devices. TMP36 Temperature Sensor Datasheet: https://www.analog.com/